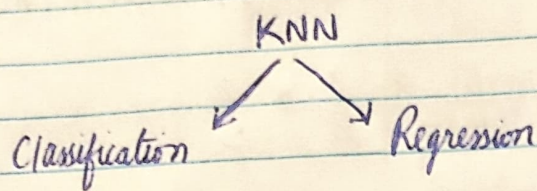
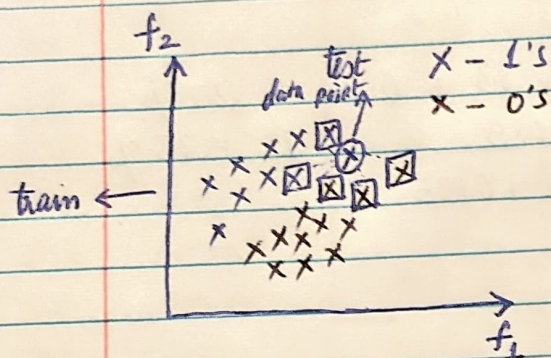


4) KNN (K-Nearest Neighbour)



Classification

f_1	f_2	O/P	} fixed no of categories
$\bar{}$	$\bar{}$	1	
		0	



K - hyperparameters

- Let's say K is 5, the value of K may be any value.

- So here the test data point will choose the 5 nearest data points since K is 5

- Maximum no of data points belongs to which category

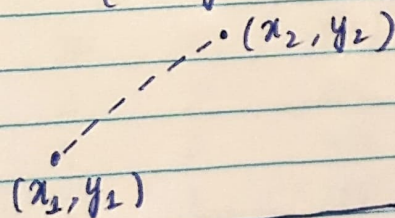
1's $\rightarrow 2$

0's $\rightarrow 3$

This is calculated using distance

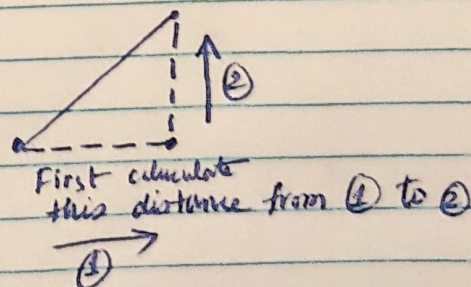
Mechanisms to calculate the distance

1) Euclidean distance
(mostly used ~~this~~ method)



$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

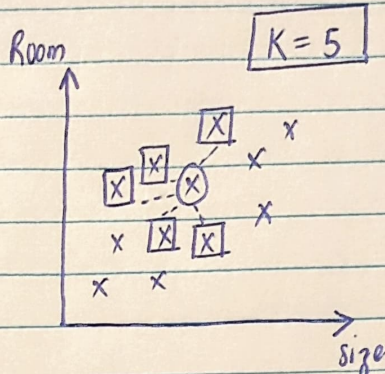
2) Manhattan Distance



Regression

{House pricing}

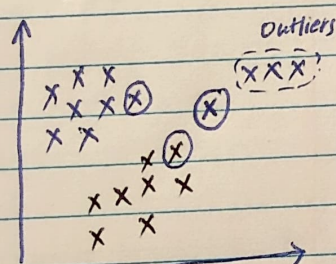
Size	room	Price
-	-	continuous value
-	-	"
-	-	"



Output $\rightarrow$ Average of this nearest data points

Limitations of KNN

- 1> Not great performance with huge data sets
- 2> it is sensitive to outliers
- 3> also sensitive to missing values



if $K=5$, then here outliers are near to the test data point